Jeremy Wang

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Interests Statistical genetics, representation learning, combinatorial algebraic geometry

Background

Ph.D. Bioinformatics 2022-Present
UCLA, Los Angeles, CA
- Ph.D. research in computational methods applied to genomic data under direction of Noah Zaitlen.

B.S. Mathematics 2018-2022
Brown University, Providence, RI
- Studied algebra, algebraic geometry, and topology. Additional coursework in algorithms and molecular biology.

Experience

Researcher 2022 - 2026
UCLA, Bioinformatics Program, Los Angeles, CA
- Advised by Noah Zaitlen
- Developed a novel statistical method that controls for heritable covariates without incurring collider bias or requiring in-sample measurements of covariates.
- Discovered novel variants affecting T2D via $\beta-$cell mediated pathways and via brain morphology and behavioral traits.
- Used modern linear algebraic techniques and GPU acceleration to enable fast, autodifferentiable heritability estimation to regularize representation learning.
- Developing a novel vision transformer-based segmentation method for 3D brain magnetic resonance imaging (MRI) using a Taylor-approximated canonical correlation objective function.

Researcher 2022 - 2022
Brown University, Mathematics Department, Providence, RI
- Advised by Melody Chan
- Computed the Frobenius characteristic of the S_n representation on labelled threshold graphs and the corresponding hyperplane arrangements, categorified a theorem of Stanley's to prove a special case of Proudfoot's equivariant log-concavity conjecture for matroids.
- Computed the S_{p+q}-equivariant Euler characteristic of the moduli space of p points in general linear position and q arbitrary points for small p, q using the Grothendieck-Lefschetz trace formula.

Teaching Assistant 2021 - 2022
Brown University, Computer Science Department, Providence, RI
- Served during the fall 2021 and spring 2022 semesters, for the courses CSCI 1810 and 1820 (Algorithmic foundations of computational molecular biology).
- Course led by Sorin Istrail.
- Held office hours, graded assignments, and wrote new curricula and problems in sequence alignment, genome assembly, phylogenetics, and clustering.

Research Fellow 2019 - 2021
UCLA, BIG Summer Program, Los Angeles, CA (Ernst Lab)

- Developed a functional genomic alignment of human and mouse genomic regions (ALLOI, ALignment by Learning Of an Immersion) using locally linear mappings of more than 2000 epigenomic assays.
- Created a novel deconvolution method for bulk ATAC-seq (ATAC-DeLFI, ATAC-seq Deconvolution by Learning Feature Importances) based on feature importances in random forest and gradient boosting models.
- Presented work in poster sessions to faculty and graduate students, winning faculty-presented awards for research excellence.

Achievements

Talks and Posters

- "Concentrating association power along specific biological pathways by controlling for heritable covariates in proxy GWAS", American Society of Human Genetics, Annual Meeting 2025
- "Deriving computationally-defined latent phenotypes from medical images and genetic data", American Society of Human Genetics, Annual Meeting 2023.
- "Subimage analysis for identifying cell-type-specific chromatin conformation features", UCLA, October 2023.
- "The Chow ring of $\overline{\mathcal{M}}_{0,n}$ by wonderful compactification", Brown MATH 2720T.
- "A functional genomic alignment of human and mouse genomic regions", UCLA BIG 2021.
- "An introduction to algebraic geometry", Brown Math DRP Spring 2021.
- "An introduction to representation theory", with Dominick Joo, Brown Math DRP Fall 2020.
- "Machine Learning Approaches for Deconvolution of Bulk ATAC-seq Signals from scATC-seq Atlases", with Rebecca Castillo, UCLA BIG 2019.

Publications

- **Jeremy Wang**, Zachary R. McCaw, Farhad Hormozdiari, and Noah Zaitlen, Accelerated, autodifferentiable variance component analysis enables genetics-optimized representation learning, In preparation.
- **Jeremy Wang** and Noah Zaitlen, Concentrating association power along specific biological pathways by controlling for heritable covariates in proxy GWAS In revision at Nature Genetics, 2026.
- Richard Border, **Jeremy Wang**... Noah Zaitlen, Simple models of non-random mating and environmental transmission bias standard human genetics statistical methods Nature Genetics, accepted in principle, 2026.
- Selena Lin, Adam Zhang, Jessica Lian, **Jeremy Wang**... Ying-Hsiu Su, Recurrent hbv integration targets as potential drivers in hepatocellular carcinoma. Cells, 10(6), p.1294. (2021).
- **Jeremy Wang**... Ying-Hsiu Su, Development and Evaluation of Novel Statistical Methods in Urine Biomarker-Based Hepatocellular Carcinoma Screening. Scientific reports, 8(1), pp.1-8. (2018).

Awards

- UCLA NIH T32 Predoctoral Trainee, 2023-2025